


Exercises.

1. Show that in the 3-dimensional complex vector space, the three vectors $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 1 \\ w \\ w^* \end{pmatrix}$, $\begin{pmatrix} 1 \\ w^* \\ w \end{pmatrix}$, where $w = e^{i\frac{2\pi}{3}}$, are orthogonal to one another. Furthermore, a vector $\begin{pmatrix} u \\ v \\ x \end{pmatrix}$ orthogonal to all three must vanish. Prove that in d -dimensional complex vector space there can be at most d mutually orthogonal vectors.

$$\begin{aligned}
 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ w \\ w^* \end{pmatrix}^\dagger &= 1 + w + w^* \\
 &= 1 + e^{-i\frac{2\pi}{3}} + e^{i\frac{2\pi}{3}} \\
 &= 1 + \cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right) + \cos\left(-\frac{2\pi}{3}\right) - i\sin\left(-\frac{2\pi}{3}\right) \\
 &= 1 + 2\cos\left(\frac{2\pi}{3}\right) \quad \parallel \quad \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} \\
 &= 1 - 1 = 0
 \end{aligned}$$

Similarly:

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ w^* \\ w \end{pmatrix}^\dagger = 1 + e^{i\frac{2\pi}{3}} + e^{-i\frac{2\pi}{3}} = 0$$

And

$$\begin{aligned}
 \begin{pmatrix} 1 \\ w \\ w^* \end{pmatrix} \begin{pmatrix} 1 \\ w^* \\ w \end{pmatrix}^\dagger &= 1 + w^2 + (w^*)^2 \\
 &= 1 + e^{i\frac{4\pi}{3}} + e^{-i\frac{4\pi}{3}} \\
 &= 1 + 2\cos\left(\frac{4\pi}{3}\right) = 1 - 1 = 0
 \end{aligned}$$

three
The $\uparrow$ vectors are orthogonal to each others!

Proof that in d -dimensional complex vector space there can at most be d mutually orthogonal vectors.

The basis vectors are orthogonal if $e_i \cdot e_j = \delta_{ij}$

any vector can be constructed from basis vectors so we need only prove this for basis vectors

Since we have d -dimensional complex vector space the indices of e_i must run from 1 to d . We can demonstrate this by using the column vector as example a d -dimensional column vector has d -rows:

$$\begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \Leftrightarrow d\text{-dimensional, this means}$$

$\sum_{i,j} e_i \cdot e_j = d \Rightarrow$ the amount of mutually orthogonal basis vector

can at most be d . This is because we define basis vector as $e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$, ..., $e_d = \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}$

so there are d -possible options at maximum $\Rightarrow d$ -mutually orthogonal iff

Orthogonality: $\langle v_i, v_j \rangle = 0$

d -dimensional vector space is spanned by d linearly independent vectors.

Orthogonality implies linear independence

$\Rightarrow$ at most d mutually orthogonal vectors

2. Determine the multiplication table of the class algebra for

$$D_5 = C_{50}$$

D_5 has 4 classes $\parallel D_5 = \{I, r, r^2, r^3, r^4, s_0, s_1, s_2, s_3, s_4\}$

class 1: $\{I\}$

class 2: $\{r, r^4\}$

class 3: $\{r^2, r^3\}$

class 4: $\{s_0, s_1, s_2, s_3, s_4\}$

The class average is defined by

$$K(c) \equiv \frac{1}{n_c} \sum_i g_i^{(c)}$$

$$\Rightarrow K(1) = I$$

$$K(2) = \frac{1}{2} (r + r^4)$$

$$K(3) = \frac{1}{2} (r^2 + r^3)$$

$$K(4) = \frac{1}{5} (s_0 + s_1 + s_2 + s_3 + s_4)$$

Let's now construct a multiplication table for class algebra

$$K(c) K(d) = \sum_e \Gamma(c, d; e) K(e)$$

$$K(1) \cdot K(2) = K(2) \quad \Rightarrow \quad K(1) \cdot K(c) = K(c)$$

$$\left(\Rightarrow \Gamma(I, d; e) = \delta^{ec} \right)$$

$$K(2) \cdot K(3) = \frac{1}{4} (r + r^4) (r^2 + r^3) = \frac{1}{4} (r^3 + r^4 + r + r^2)$$

$$\Rightarrow K(2) K(3) = \frac{1}{4} (r + r^4) + \frac{1}{4} (r^2 + r^3) = \frac{1}{2} (K(2) + K(3))$$

$$K(2) K(4) = \frac{1}{10} (r + r^4) (s_0 + s_1 + s_2 + s_3 + s_4)$$

$$= \frac{1}{10} r (s_0 + s_1 + s_2 + s_3 + s_4) + \frac{1}{10} r^4 (s_0 + s_1 + s_2 + s_3 + s_4)$$

$$= \frac{1}{10} (s_1 + s_2 + s_3 + s_4 + s_0) + \frac{1}{10} (s_4 + s_0 + s_1 + s_3)$$

$$K(2) K(4) = \frac{1}{2} K(4) + \frac{1}{2} K(4) = K(4)$$

Similarly

$$K(3) \cdot K(4) = \frac{1}{2} K(4) + \frac{1}{2} K(4) = K(4)$$

Now we just need $K(2)K(2)$, $K(3)K(3)$ & $K(4)K(4)$

$$\Rightarrow K(2)K(2) = \frac{1}{4} (r + r^4) (r + r^4) = \frac{1}{4} (r^2 + I + I + r^3)$$

$$= \frac{1}{2} K(1) + \frac{1}{2} K(3)$$

$$K(3)K(3) = \frac{1}{4} (r^2 + r^3) (r^2 + r^3) = \frac{1}{4} (r^4 + I + I + r)$$

$$= \frac{1}{2} K(1) + \frac{1}{2} K(2)$$

$$K(4)K(4) = \frac{1}{25} (s_0 + s_1 + s_2 + s_3 + s_4) (s_0 + s_1 + s_2 + s_3 + s_4)$$

$$= \frac{1}{25} (I + r^4 + r^3 + r^2 + r) \cdot 5 = \frac{1}{5} (I + r + r^4 + r^2 + r^3)$$

$$= \frac{1}{5} K(1) + \frac{2}{5} K(2) + \frac{2}{5} K(3)$$

Using these we can form the multiplication table.

Multiplication table of the class algebra for $D_8 = C_{5v}$

	$K(1)$	$K(2)$	$K(3)$	$K(4)$
$K(1)$	$K(1)$	$K(2)$	$K(3)$	$K(4)$
$K(2)$	$K(2)$	$\frac{1}{2}(K(1) + K(3))$	$\frac{1}{2}(K(2) + K(4))$	$K(4)$
$K(3)$	$K(3)$	$\frac{1}{2}(K(2) + K(4))$	$\frac{1}{2}(K(1) + K(3))$	$K(4)$
$K(4)$	$K(4)$	$K(4)$	$K(4)$	$\frac{1}{5}(K(1) + 2K(2) + 2K(3))$

3 Show that $f(c, d, I) = \delta_{\bar{c}d}/n_c$.

$\bar{c}$ is inverse element of c

$$K(I) K(d) = \sum_e n_e f(I, d, e) K(\bar{e})$$

$$K(d) = \sum_e n_e f(I, d, e) K(\bar{e}) \Rightarrow f(I, d, e) = \frac{1}{n_c} \int^{d\bar{e}} K(\bar{e})$$

$$\Rightarrow f(c, d, \pm) = \delta_{\bar{c}d}/n_c$$

4. Show that $f(c, d, e) = f(\bar{c}, \bar{d}, \bar{e})$

$$K(c) K(d) = \sum_e n_e f(c, d, e) K(\bar{e})$$

$$K(c) K(d) K(e) = \sum_a n_a f(c, d, a) K(\bar{a}) K(e) = f(c, d, e) I + \dots$$

$$K(a) K(b) = \frac{1}{n_a n_b} \sum_{ij} g_i^{(a)} g_j^{(b)} = \sum_e n_e f(a, b, e) K(\bar{e})$$

$$K(\bar{a}) K(\bar{b}) = \frac{1}{n_a n_b} \sum_{ij} (g_i^{(a)} g_j^{(b)})^{-1} = \sum_e n_e f(a, b, e) K(e)$$

because $(g_i^{(a)} g_j^{(b)})^{-1}$ is invm of $g_i^{(a)} g_j^{(b)}$ so instad of $K(\bar{e})$ we get $K(e)$
 the coeff $f(a, b, e)$ is same since it only measures how many times
 term of $(g_i^{(a)} g_j^{(b)})^{-1}$ or $(g_i^{(a)} g_j^{(b)})$ apper i.e $K(\bar{e})$ or $K(e)$ appers

We can also write

$$K(\bar{a}) K(\bar{b}) = \sum_e n_e f(\bar{a}, \bar{b}, e) K(\bar{e}) \quad \text{with handhold upgrade}$$

$$\Rightarrow K(\bar{a}) K(\bar{b}) = \sum_e n_e f(a, b, e) K(e) = \sum_e n_e f(\bar{a}, \bar{b}, e) K(\bar{e})$$

if we add one more component

$$\begin{aligned} \Rightarrow K(\bar{a}) K(\bar{b}) K(\bar{c}) &= \sum_e n_e f(a, b, e) K(e) K(\bar{c}) = f(a, b, c) I + \dots \\ &= \sum_e n_e f(\bar{a}, \bar{b}, e) K(\bar{e}) K(\bar{c}) = f(\bar{a}, \bar{b}, \bar{c}) I + \dots \end{aligned}$$

$$\Rightarrow f(a, b, c) = f(\bar{a}, \bar{b}, \bar{c}) \quad \text{QED}$$

Warum wichtig -

5. Prove (3b)

$$K(c) K(d) K(e) = (K(c) K(d)) K(e) = K(c) (K(d) K(e))$$

$$\Rightarrow \sum_a n_a f(c, d, a) K(\bar{a}) K(e) = \sum_a n_a f(d, e, a) K(\bar{a}) K(c)$$

$$\Rightarrow \sum_{ab} n_a n_b f(c, d, a) f(\bar{a}, e, b) K(\bar{b}) = \sum_{ab} n_a n_b f(d, e, a) f(\bar{a}, c, b) K(\bar{b}) \quad || \text{rename } a \leftrightarrow e$$

$$\sum_{eb} n_e n_b f(c, d, e) f(\bar{e}, a, b) K(\bar{b}) = \sum_{eb} n_e n_b f(d, a, e) f(\bar{e}, c, b) K(\bar{b})$$

$$\Rightarrow \sum_e n_e f(c, d, e) f(\bar{e}, a, b) = \sum_e n_e f(d, a, e) f(\bar{e}, c, b)$$

$$\Rightarrow \boxed{\sum_e f(c, d, e) f(\bar{e}, a, b) = \sum_e f(d, a, e) f(\bar{e}, c, b)} \quad Q.E.D$$

proved!!

6.

6 Use Schur's lemma to prove the almost self-evident fact that all irreducible representations of an abelian group are 1-dimensional.

Schur's lemma: if $D(g)$ is an irreducible representation of finite group G & if there is some matrix A such that $A D(g) = D(g) A$ for all g , then $A = \lambda I$ for some constant λ .

For abelian groups we have $D(g_1) D(g_2) = D(g_1 g_2) = D(g_2 g_1) = D(g_2) D(g_1)$

$$D(g_1) D(g_2) = D(g_2) D(g_1)$$

meaning $D(g_2) = \lambda_2 I$ & same for $D(g_1)$

$$\text{i.e. } D(g_1) = \lambda_1 I$$

$\Rightarrow D(g) \in \{\lambda_i\} \cdot I$ i.e. only λ_i is needed to

differentiate between $D(g)$'s meaning $D(g)$ is one dimensional.
or $I = 1 \times 1$